

# PHA4GE Wastewater Contextual Data SOP

Version 4.3 Updated April 2026

## I. Purpose

The purpose of this SOP is to structure wastewater contextual data according to the PHA4GE contextual data collection templates in order to better enable harmonization across datasets and systems. Standardising contextual information ensures data is interoperable, shareable, and reusable while also protecting privacy and supporting ethical data use.

- A. Data providers will curate contextual data according to the steps outlined in the procedure below.
- B. Data providers will populate the appropriate template with information from their datasets using applicable picklists and according to instructions.
- C. Data providers can share the harmonized data according to jurisdictional and organization-specific data sharing policies.

This SOP and the supporting materials are available on GitHub at <https://github.com/pha4ge/wastewater-contextual-data-specification>.

## II. Data

The scope of the standard encompasses contextual data describing wastewater samples, including their collection context, environmental and infrastructure characteristics, upstream population attributes, and laboratory processing and sequencing metadata, such that the resulting data are interoperable, comparable, and semantically well-defined in accordance with PHA4GE wastewater contextual data specification best practices.

**Note:** Different subsets of fields will apply to samples from different contexts (i.e., structured sewage systems vs. contaminated surface waters). Fields not pertinent to the sample type of interest need not be filled. Refer to the Appendix B and the Reference Guide for further instructions regarding different sample types.

## III. Key Principles

Use of a standardised template allows for consistent, comparable datasets with easier integration across sites and systems, while also future-proofing for reuse. Collecting contextual data in a standardized way also enables better protection of sensitive data. Consider the following when completing the contextual data:

- II. **Permission to Share:** Data providers must have the rights and permissions from the data owners to submit and share the contextual data. Ensure alignment with institutional, jurisdictional, and legal data sharing and privacy policies.

- III. **Privacy and Ethics:** Respect ethical constraints. Use null values when data is not permitted for sharing. Seek guidance from a data steward or privacy officer if needed.
- IV. **Attribution:** Ensure all contributors (sample collection, processing, sequencing, curation) are credited.

## IV. Procedure

### 1. Preparation

- Download the file containing the current iMicroSeq reference guide from the GitHub repository [https://github.com/pha4ge/Wastewater\\_Contextual\\_Data\\_Specification](https://github.com/pha4ge/Wastewater_Contextual_Data_Specification).
- To use the DataHarmonizer as your data collection instrument follow the instructions for [using the DataHarmonizer](#). You can download the latest wastewater DataHarmonizer templates at the [pathogen genomics package](#).
- Prior to curating contextual data please:
  - Confirm data sharing permissions with your institution, considering privacy, legal, and ethical constraints.
  - Review your dataset accordingly
  - Review the fields and terms in the Reference Guide (schema).
  - Review the field definitions and guidance in the template Reference Guide
  - Confirm the planned mapping of your data fields to those in the PHA4GE collection template with the data steward (e.g. your supervisor).
  - *Note: Confirm the level of granularity of information that can be shared publicly and/or privately, with the data steward and/or your privacy officer. The most detailed information allowable should be included here. Different versions (detailed information vs general information) can be stored.*

### 2. Completing the Template

- Populate the collection template with information from your dataset. You can use the DataHarmonizer for a submission ready template to downstream repositories, or alternatively you can use the reference guide to create your own template.
  - Fill in the **required fields** (highlighted yellow in the reference guide, alternatively see the table below).
  - Complete **recommended fields** (highlighted purple in the reference guide, alternatively see the table below) where possible.
  - Complete optional fields (white) if data is available and approved for sharing.
  - Use controlled vocabularies and picklists.
  - Use null values appropriately to indicate reasons for missing data (e.g., not collected, not applicable, restricted access).
  - Ensure the data is stored safely with appropriate encryption.
- Use the wastewater contextual data Reference Guide to access field definitions, field-level guidance and examples.

*Note: Sometimes there will be constraints on what information can be shared, other times a field may not be applicable to your sample. Use the null values (controlled vocabulary indicating the reason why information is not provided) in the picklist to report missing data. (an asterisk “\*” denotes exceptions between templates)*

Table: Summary of the required (bold) and recommended fields of the PHA4GE Wastewater contextual data specification.

| Subsection                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Required Fields                                                                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Database Identifiers</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                     | specimen collector sample ID                                                                                                                                                           |
| <b>Sample Collection and Processing</b><br><i>Note: Consult your supervisor and/or data steward to evaluate whether the specimen collector sample ID is considered identifiable according to your institutional policies. If not considered identifiable, copy the sample ID into the “specimen collector sample ID” field in the collection template. If considered identifiable, provide the alternative sample ID. Be sure to keep a copy of the key in a safe location.</i> | sample collected by<br>geo_loc_name (country)<br>geo_loc_name (state/province/territory)<br>organism* (not required for AMR template)<br>purpose of sampling<br>sample collection date |
| <b>Strain and isolate information</b>                                                                                                                                                                                                                                                                                                                                                                                                                                           | isolate ID* (only required for Pathogen Agnostic template)                                                                                                                             |
| <b>Sequence Information</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                     | purpose of sequencing<br>sequenced by<br>sequenced by contact name<br>sequenced by contact email<br>sequencing instrument                                                              |
| <b>Taxonomic identification information</b>                                                                                                                                                                                                                                                                                                                                                                                                                                     | read mapping software name* (only required for Pathogen Agnostic template)<br>read mapping software version* (only required for Pathogen Agnostic template)                            |

### 3. Data Handling and Storage

If IDs or identifiers are anonymised, securely retain the mapping key.

## V. Ethical and Practical Considerations

Effective and equitable pathogen surveillance via wastewater sampling requires rapid and sustained international collaboration and data sharing. Many of the contextual data elements described in the PHA4GE wastewater contextual data specification are critical for effective

public health surveillance and response. However, many of these same elements have ethical, practical, and privacy issues which must be considered before data can be shared. Data governance policies may vary between data types and jurisdictions, thus users of the specification should consult data stewards and privacy officers regarding organization-specific and jurisdiction-specific policies. At the highest level this includes:

- **Granularity:** Share the most granular information allowable, balancing utility with privacy.
- **Geolocation and Temporal Data:** Use accurate and approved spatial/temporal data. Avoid using proxy locations.
- **Methodological Transparency:** Record relevant sampling, processing, and analysis methods. These support data reuse and reproducibility.
- **Licensing and Use:** If applicable, include data use license information.

Below, we highlight a series of common issues and provide suggestions for ways forward. The PHA4GE Reference Guide should be consulted for field-level guidance.

*Note: This guidance is based on the experience of members of the PHA4GE working groups, and is not intended to apply to all situations and use cases. Decisions regarding implementation of the specification must ultimately be made by the user in consultation with data providers and data stewards. If the intended use of the information collected is for research purposes, there will likely be many additional administrative and ethical requirements (e.g. Research Ethics Board (REB) review).*

## 1. Data Stewardship and Attribution

Data is often generated through the work and contributions of different individuals and organisations (e.g. experimental design, sample collection, microbial isolation, sequencing, laboratory testing/characterization, etc). Attributing the contributions from different partners can be critical for ensuring trust within collaborations, sharing equitable distribution of the benefits (e.g. establishing track records, applying for grants), and for providing accurate contact details for follow-up. As such, data provenance fields such as *sample collected by organisation name*, *isolated by organisation name*, and *sequenced by organisation name* should be filled when possible. Formal organization names (what might appear on official letterhead) should be provided instead of short forms e.g. “University of British Columbia” rather than “UBC”. The *authors* field enables general attribution of different partners involved in data generation, and has no limit.

Data stewardship is the management, protection, and oversight of data assets to ensure quality, security, and accessibility throughout their lifecycle. Individuals can be nominated as the data steward of datasets submitted to the Portal. Data stewards will be responsible for long-term oversight of data, and should have contact information available for follow-up (e.g. *data steward contact name* and *data steward contact email* are required fields).

## 2. Identifiers and Repository Accession Numbers

Laboratories world-wide are collecting and analyzing wastewater data; however, while it is common to share clinical sequence and minimal contextual data with public repositories such as GISAID and the INSDC, this is not yet a common practice following wastewater surveillance. Through sharing consensus sequence and raw data, as well as contextual data, with public repositories, we can:

- Detect diverse pathogens from viral, bacterial, and other backgrounds
- Guide vaccination campaigns (ex: polio)
- Develop and improve diagnostic technologies for wastewater samples
- Supplement clinical surveillance in order to understand community rates of infection and predict hospitalization rates
- Inform investment in water sanitation and hygiene

When you share information with a public database, you will receive an accession number (a unique identifier in a database enabling the tracking of multiple versions of the data). If you have shared data with a public database, make sure to capture the accession numbers. GISAID will provide you with a single accession number. Make sure to record it. INSDC members (NCBI, ENA, DDBJ) may provide you with different accession numbers depending on what you share, and how. You can share assemblies and consensus sequences with GenBank (and its equivalents), raw data with Sequence Read Archive (SRA), and contextual data as a BioSample (see reference guide for further information). Information may be organized in BioProjects, and at a higher organizational level, Umbrella BioProjects. Make sure to record all of the applicable accession numbers. Samples, libraries, sites, sequences (raw, processed, consensus etc) and so on can have many identifiers, especially if there is a division of labor or sharing of information across agencies and organizations. The specification has provided fields to capture many of those that are common, but may not capture all of the IDs you require. **It is essential to track IDs of original materials and information** to establish chain-of-custody and for follow-up, if necessary. It is better to track too many IDs than too few. If you require more fields to capture the IDs you need, add them. Some IDs are considered public health identifiable information (PHII). Make sure to check with the appropriate authorities whether the IDs you plan to share are considered identifiable information. If considered identifiable, you may need to create an alternative set of IDs. If you do, make sure to store the key in a safe and secure place.

## 3. Geographical information

Geographical information (country, province/state/region, city, postal code, latitude/longitude etc) is an essential descriptor of a wastewater site. However, when coupled with other metadata, this information may incidentally reveal the identities of individuals or a community contributing effluent to a given site. Especially for sites with smaller catchment areas, it may be necessary to abstract geographic information before it can be shared. Before sharing data, especially with public repositories, it is important to ensure the data being submitted complies with the permitted level of granularity. Discuss this with the data steward. If sharing latitude and longitude

coordinates, do not use the center of the region or the location of your agency as a proxy, as this implicates a real location and is misleading. Latitude and longitude should either be shared accurately or kept private.

Sewershed/watershed information can also potentially be used to identify a specific community contributing effluent to a site. For example, if a site is described as a “correctional facility,” this could be combined with geographic information to positively identify a specific prison. Proximal sites can also make data identifiable. Especially when a community is smaller or disadvantaged, the potential to identify a specific community should be taken into account when deciding which contextual data to share with collaborators or publicly.

#### 4. Date information

Geographical and temporal information are key elements of infectious disease surveillance programs. Temporal information consists of dates e.g. sample collection date, sample received date, sample sequenced date, etc. This information should be captured using clearly defined event-specific fields (e.g. sample collection date, sample received date, sequencing date), with all dates recorded in ISO 8601 format (YYYY-MM-DD) to ensure consistency and interoperability.

The “sample collection date” represents the time at which the specimen was obtained from its source and should not be inferred or substituted. The “received date” must only be used to record when the sample arrived at the processing laboratory and must not be used as a proxy for collection date, as this introduces systematic bias in temporal analyses. Elements such as “sample collection date” are usually held by the institution that collected the original specimen. As such, you may require permission to acquire this information, or it may be difficult to attain due to other burdens on the data provider (workload, system access, manual curation requirements). If the sample collection date is unavailable, the appropriate null value should be used rather than substituting another date field. This preserves semantic clarity and prevents misinterpretation in downstream analyses.

#### 5. Purpose of sampling/sequencing

Sampling strategy can create biases in the data. A sample may be collected for one purpose, but sequenced for another (e.g. collected for diagnostic testing, but sequenced for surveillance of circulating lineages and variants). Information about why samples were collected and why they were selected for sequencing (i.e. random vs targeted sampling) can help inform epidemiological modeling and analyses. Standardized tags are available in the “purpose of sampling” field (e.g. protocol testing, wastewater drug surveillance) and in the “purpose of sequencing” field (e.g. baseline surveillance (non-random sampling), travel-associated surveillance). Free text fields are also available for providing extra information about sampling and the selection of samples for sequencing called “purpose of sampling details” and “purpose of sequencing details”. A number of standardized phrases are also suggested in the Reference Guide for describing different common surveillance priorities.

## 6. Methods information

Methodological information, such as sampling and experimental design, laboratory procedures, bioinformatic processing, and quality control metrics, are crucial information to understand the context and limitations of analyses. Capturing as much well-structured information regarding your methods, and storing it in a centralized place (or single document) helps to futureproof the data as well as the work that went into collecting, processing, analyzing and interpreting the data. Capturing methodological information also enables better reproducibility, and increases quality control. The specification provides many fields for capturing experimental design, protocols, and scientific metrics. It is strongly recommended that as much of that information be captured and stored as possible. It should be noted that the template, while designed to capture critical information about wastewater surveillance, **is not meant to function in the same way as a Laboratory Information Management System (LIMS)**. Far more detailed methods may be included in LIMS (e.g. concentrations, values, measurements used for preparing libraries). The PHA4GE templates serve as a summary of this information. However, in the absence of a formal LIMS, these templates can be used for tracking and storing methodological information.

When completing **bioinformatics fields**, only those relevant to your workflow should be populated. Some fields may appear similar, such as consensus sequence software name and sequence assembly software name, but they represent distinct methods of analysis. The sequence assembly software field should be completed if your workflow includes an assembly step (for example, de novo assembly), while the consensus sequence software field should be completed if your workflow generates consensus sequences from mapped reads. If both steps occur, both fields should be populated. In cases of uncertainty regarding which fields to complete, the data steward should be consulted prior to submission.

Similarly, when completing the taxonomic information section the relevant fields should be completed if a taxonomic analysis has been conducted. This includes recording the read mapping software name and version, the taxonomic reference database name and version, and the read mapping criteria applied. Any generated outputs, such as a taxonomic analysis report, should be referenced by filename and associated with the date the analysis was performed. If you are unclear about which fields to use, contact the data steward for clarification.

## 7. Recording measurements: Exact values vs bins

While it is ideal for measurements to be as precise as possible for comparisons across studies, sometimes only ranges of values are available. The templates are designed to accommodate both exact and binned measurements, allowing for flexible data capture across laboratories with varying public health priorities, capacities, and reporting needs. Exact values and binned ranges should be recorded in their respective fields (e.g., *water catchment area human population measurement value* vs *water catchment area human population range*).

With the exception of individual count measurements, all measurements should follow a structured value, unit, and method pattern of capture where:

- **Value** represents the quantitative observation (either a precise number or a defined bin), forming the core data used in analysis.
- **Unit** specifies the scale and standard of measurement (e.g. CFU/mL, copies/L), enabling harmonisation, comparability, and correct interpretation across datasets.
- **Method** describes how the measurement was generated (e.g. assay type, instrument, protocol), providing essential context for assessing data quality, comparability, and potential bias.

Capturing these components in separate but linked fields preserves semantic clarity, supports ontology alignment, and enables computational reuse.

## 8. Attribution

The generation of genomic sequences from samples requires a lot of work, and can involve contributions from different partners - from the acquisition of funding, to sample collection, to sample processing and sequencing, to analysis and interpretation, to sequence and contextual data management and submission. There are many roles and tasks along the genomic surveillance continuum and it is important to track and acknowledge the contributions of all those involved. Attribution of contributions can be critical for demonstrating productivity for securing future funding, for establishing chains of custody, for ensuring contact information is available for follow-up, and is essential for ethical benefit sharing in partnerships. Discussions regarding how individuals and/or organizations should be attributed in collections (private/public), manuscripts and other published or non-published work, should be discussed prior to beginning any project. The “authors” field in the template can be populated with a list of contributors as well as short descriptions of their contributions. Data stewards may want to consider the inclusion of data license tags (fields and values denoting data use permissions/restrictions) in their records (note: data license tags not included in the templates).

## 9. Multi-Tagging

Users may find that multiple values are necessary in some fields. The templates provide “multi-tagging” capability enabling users to enter multiple values separated by a semi-colon. Users who create their own templates from the PHA4GE master list of vocabulary and enable multi-tagging should also use semicolons as separators as this is the convention preferred by large public repositories.

## 10. Null Values

The International Nucleotide Database Collaboration (INSDC) have created standardized missing/null value reporting language to be used where a value of an expected format for sample metadata reporting can not be provided. This controlled vocabulary has been adopted in this specification, and takes into account different types of constraints (i.e. Not Applicable, Missing, Not Collected, Not Provided, Restricted Access). Users are strongly encouraged to always provide as much information as possible in the collection template, however, if missing/null value reporting is required, users are asked to use a term with the finest granularity



for their situation. Note: NCBI accepts all null values. ENA will accept any other null value besides "Missing".

See **Appendix A** for information on use case specific templates. See **Appendix B** for examples of how to structure sample descriptions.

If a desired term is not present in a picklist, you can request standardized terms using the procedure in **Appendix C**.

## Appendix

### Appendix A: Use Case-specific Templates

Wastewater genomic surveillance data can have many applications, and be used to explore data in different ways in order to answer a wide variety of public health questions. Often, the data that is collected is shaped by these priorities, and the questions that an organization is trying to answer. As such, contextual datasets can vary in content, granularity, and structure. To address this variability in data collection, to provide as much flexibility as possible, and to reduce unnecessary data entry, the PHA4GE Wastewater Specification package contains different templates scoped for different widely used use-cases. These use-cases include wastewater-based SARS-CoV-2 surveillance (SARS-CoV-2 template); wastewater-based antimicrobial resistance detection, surveillance, and research (AMR template); and detection and surveillance of other target pathogens (Pathogen Agnostic template). The templates are scoped for capture of contextual information used in amplicon-based sequencing, single isolate whole genome sequencing, as well as metagenome sequencing. The specification package captures information about sequence reads, consensus sequences and assemblies from cultured isolates, selected results from analyses (e.g. taxonomic identification of pathogens present in a wastewater sample), and the names of reports where further results can be obtained. The content of the different templates overlap, but there are important nuanced differences. **Users should identify the most appropriate template for their data. Users can also create their own spreadsheets and templates by recombining selected fields, terms and modules from the PHA4GE specification according to their needs.** A JSON file containing all possible vocabulary (the “master wastewater specification”) is also available if the user would prefer to create their own custom template from the master specification - this is encouraged for bespoke applications. More templates may be added over time, and requests can be suggested by contacting [datastructures@pha4ge.org](mailto:datastructures@pha4ge.org).

### Appendix B: Describing your sample

Why, how and when samples are collected can impact analyses of sequence data. In determining how a virus spreads, it is critical to track temporal and geographical information. It is also important to capture as much data provenance (who contributed it, where it came from, how it was generated) as possible. Different sampled materials or sampling processes may contain higher viral loads or produce better results, and differences in sampling protocols and practices should be accounted for (e.g. to understand sampling effects on the ability to identify specific pathogens).

A number of recommended and optional fields are provided to capture sampling approach (“purpose of sampling”, “scope of sampling”), collection methods (“sample collection time duration value”, “sample volume measurement value”, “collection device”, “collection method”), sample storage (“sample storage medium”, “sample storage duration value”), and sewershed details (“environmental site”, “environmental material”, “wastewater system type”). **Populate only the fields that pertain to your sample.** Provide the most granular information allowable

according to your organization's data sharing policies. **Note: only include facts that are provided in the sample description, do not make assumptions and add details not provided by the data generator.**

**e.g., a composite sample made of multiple grab samples** should be recorded:

|                     |                   |                           |
|---------------------|-------------------|---------------------------|
| Specimen processing | Collection device | Collection method         |
| Samples pooled      | Grab sampler      | Manual composite sampling |

**e.g., a 12-hr 1-liter flow-proportional composite sample** should be recorded:

|                                       |                                      |                                 |                                |                                     |                    |
|---------------------------------------|--------------------------------------|---------------------------------|--------------------------------|-------------------------------------|--------------------|
| Sample collection time duration value | Sample collection time duration unit | Sample volume measurement value | Sample volume measurement unit | Collection device                   | Collection method  |
| 12                                    | Hour                                 | 1                               | Liter (L)                      | Automatic flow-proportional sampler | Composite sampling |

**e.g., a treated sample collected from a WWTP that receives wastewater and run-off** should be recorded:

|                             |                                   |                            |                        |
|-----------------------------|-----------------------------------|----------------------------|------------------------|
| Environmental material      | Environmental material properties | Environmental site         | Wastewater system type |
| Wastewater; Run-off (water) | Treated                           | Wastewater treatment plant | Combined sewer system  |

**e.g., a sample collected from a contaminated lake** should be recorded:

|                    |                        |                                   |
|--------------------|------------------------|-----------------------------------|
| Environmental site | Environmental material | Environmental material properties |
| Lake               | Surface water          | Contaminated                      |

**e.g., a sample collected from a school latrine** should be recorded:

|                        |                    |                        |
|------------------------|--------------------|------------------------|
| Environmental material | Environmental site | Wastewater system type |
| Wastewater             | School             | Latrine                |

## Appendix C: New Field/Term Requests

Good data standards are continually maintained, and evolve over time as data needs change. In the event that the template you are using does not contain appropriate vocabulary for your needs, a new term request (NTR) can be made via two different mechanisms. Ideally, an NTR should be made via the PHA4GE Wastewater Specification GitHub Issue Tracker. Users will require a GitHub account, and should follow the instructions provided in the [New Term Request SOP](#). NTR's will require the desired field/term name, suggested definition, definition source (citation or URL for website, article, textbook, or the name of the curator suggesting the definition etc). If possible, the user should suggest the target ontology and parent class of the term (but this is not required). Alternatively, NTRs can be made by emailing [datastructures@pha4ge.org](mailto:datastructures@pha4ge.org). For more information and/or assistance, contact [datastructures@pha4ge.org](mailto:datastructures@pha4ge.org).

### Revision History

| Version | Date             | Author                         | Description of Change                                                                                                                       |
|---------|------------------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 0.0     | August 15, 2023  | Jillian Paull & Emma Griffiths | Created protocol                                                                                                                            |
| 0.1     | December 12 2023 | Emma Griffiths                 | Removed instructions for identifying standardized terms (using OLS)                                                                         |
| 0.2     | January 17 2024  | Emma Griffiths                 | Added new sections (Templates, Attribution, Metrics, added required fields), updated methods section, updated new term request instructions |
| 0.3     | February 09 2024 | Charlie Barclay                | Updated links to DH SOP.                                                                                                                    |
| 1.0     | March 22 2024    | Emma Griffiths                 | Changed field requirements, expanded template descriptions, and added guidance on multi-tagging.                                            |
| 3.1     | October 07 2024  | Charlie Barclay                | Version compatibility update                                                                                                                |
| 4.3     | April 28 2026    | Charlie Barclay                | Update formatting to                                                                                                                        |

|  |  |  |                                                                                                                                    |
|--|--|--|------------------------------------------------------------------------------------------------------------------------------------|
|  |  |  | <p>match iMicroSeq for consistency.</p> <p>Additional information for Data (scope) and Key Principles sections. Updated links.</p> |
|--|--|--|------------------------------------------------------------------------------------------------------------------------------------|